

Competitive Programming

Section 41 Editorial: Euler Tour, LCA, and Binary Lifting

Competitive Programming Course

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1 Editorial overview

1.1 Problem-specific proof map

A. Tree Distances. Euler entry/exit times make a subtree a contiguous interval, while binary lifting raises two vertices to their lowest common ancestor. The identity $\text{dist}(u,v) = \text{depth}[u] + \text{depth}[v] - 2 * \text{depth}[\text{lca}(u,v)]$ is exact because the shared root-to-LCA prefix is counted twice. Preprocessing costs $O(n \log n)$ and each query costs $O(\log n)$.

B. Max Edge On Path. Alongside $\text{up}[j][v]$, store the maximum edge on the 2^j -edge climb. Combining the same climbs used by LCA covers the two disjoint path halves, so their maximum is the answer. Queries remain $O(\log n)$.

C. K-th Node On Path. The path is the upward segment from u to the LCA followed by the reverse upward segment from v . Compare k with the first segment length and lift either endpoint by the corresponding number of edges; this covers each path vertex once.

1.2 Detailed solution guide

A. Tree Distances. Root the tree once and compute $\text{depth}[v]$ and $\text{up}[k][v]$, the 2^k -th ancestor of v . To answer $\text{lca}(u,v)$, first lift the deeper vertex until both depths match. Then lift both vertices from high powers down to low powers whenever their ancestors differ. Their parents are then the LCA. The distance formula is $\text{depth}[u] + \text{depth}[v] - 2 * \text{depth}[\text{lca}]$.

B. Maximum Edge on a Path. Store one more table: $\text{mx}[k][v]$, the maximum edge weight on the path from v up by 2^k edges. When lifting a vertex, merge the corresponding mx value into the answer. The path from u to v splits into $u \rightarrow \text{lca}$ and $v \rightarrow \text{lca}$, so lifting both sides and taking the maximum over the chunks covers every path edge exactly once.

C. k-th Node on a Path. Let $w = \text{lca}(u,v)$. The path length in vertices is $\text{depth}[u] - \text{depth}[w] + 1 + \text{depth}[v] - \text{depth}[w]$. If k lies on the u to w part, jump $k-1$ steps upward from u . Otherwise the answer lies on the w to v part; convert it to a jump upward from v by counting how many vertices remain after position k .

Common pitfalls. Binary lifting uses powers of two, so allocate enough levels for the maximum n . For C, be explicit about whether k is 1-based; the local statement uses 1-based path positions.

This section has one preprocessing object: a binary-lifting table rooted at node 1. The three exercises reuse it with different query payloads.

- A uses only depth and up to compute $\text{lca}(u,v)$, then applies the distance formula.
- B stores $\text{mx}[k][v]$, the maximum edge on the upward jump of length 2^k . Every time the query lifts a vertex, it merges the corresponding maximum.
- C first computes the LCA and the total path length in nodes. If the requested k stays on the u side, jump upward from u ; otherwise convert it to a jump upward from v .

The proof pattern is the same in all three: the path is decomposed into disjoint upward power-of-two chunks, and the table stores the exact ancestor or aggregate for each such chunk.

2 A. Tree Distances

2.1 Restatement

Given a rooted tree and queries u, v , print the number of edges on the path.

2.2 Observations, trace, and algorithm

For n, q near $2e5$, parent climbing fails on a chain. Store $up[j][v]$, the 2^j ancestor, and depth. For sample nodes 4 and 5, $lca=2$, hence distance is $2+2-2*1=2$. Lift the deeper node, then descend powers from largest to zero, lifting both only when their ancestors differ; return their parent.

2.3 Correctness proof and complexity

$up[j][v]=up[j-1][up[j-1][v]]$ composes two exact half-jumps. Reverse-bit lifting leaves both vertices immediately below their lowest common ancestor, so their parent is the LCA. The two root paths are disjoint below it, proving the distance formula. Preprocessing is $O(n \log n)$, query time $O(\log n)$, memory $O(n \log n)$. QED.

2.4 Full C++ solution

```
#include <bits/stdc++.h>
using namespace std;
int main(){ios::sync_with_stdio(false);cin.tie(nullptr);
    int n,q;if(!(cin>>n>>q)) return 0; int LOG=1; while((1<<LOG)<=n+1) LOG++;
    vector<vector<int>> up(LOG, vector<int>(n+1)); vector<int> depth(n+1);
    for(int v=2;v<=n;v++){cin>>up[0][v]; depth[v]=depth[up[0][v]]+1;}
    for(int k=1;k<LOG;k++) for(int v=1;v<=n;v++) up[k][v]=up[k-1][up[k-1][v]];
    auto lift=[&](int v,int d){for(int k=0;k<LOG;k++) if(d&(1<<k)) v=up[k][v]; return v;};
    auto lca=[&](int a,int b){if(depth[a]<depth[b]) swap(a,b); a=lift(a,depth[a]-depth[b]); if
    while(q--){int a,b;cin>>a>>b; int c=lca(a,b); cout<<depth[a]+depth[b]-2*depth[c]<<'\n';}
}
```

2.5 Full Python solution

```

import sys

def main():
    data=list(map(int,sys.stdin.buffer.read().split()))
    if not data: return
    n,q=data[0],data[1]; idx=2
    LOG=max(1,(n+1).bit_length())
    up=[[0]*(n+1) for _ in range(LOG)]; depth=[0]*(n+1)
    for v in range(2,n+1):
        p=data[idx]; idx+=1; up[0][v]=p; depth[v]=depth[p]+1
    for k in range(1,LOG):
        for v in range(1,n+1): up[k][v]=up[k-1][up[k-1][v]]
    def lift(v,d):
        bit=0
        while d:
            if d&1: v=up[bit][v]
            d//=2; bit+=1
        return v
    def lca(a,b):
        if depth[a]<depth[b]: a,b=b,a
        a=lift(a,depth[a]-depth[b])
        if a==b: return a
        for k in range(LOG-1,-1,-1):
            if up[k][a]!=up[k][b]: a=up[k][a]; b=up[k][b]
        return up[0][a]
    out=[]
    for _ in range(q):
        a,b=data[idx],data[idx+1]; idx+=2; c=lca(a,b)
        out.append(str(depth[a]+depth[b]-2*depth[c]))
    print("\n".join(out))
if __name__=="__main__": main()

```

2.6 Tests that matter

Test $u=v$, root queries, ancestors, a chain, a maximum edge at the LCA boundary, both path endpoints, and invalid k values.

3 B. Max Edge On Path

3.1 Restatement

Each edge to a parent has a weight. For each $u \ v$, print the maximum edge weight on the path.

3.2 Observations, trace, and algorithm

Carry $mx[j][v]$, the maximum edge on the same 2^j upward segment. Whenever a lift uses bit j , merge $mx[j][v]$ before moving v . Its recurrence is $\max(mx[j-1][v], mx[j-1][up[j-1][v]])$; merge both sides while finding LCA.

3.3 Correctness proof and complexity

The recurrence joins exactly adjacent half-segments. Lifting partitions the $u-v$ path into those segments, so their merged maximum is exact. Costs match A; the empty local path $u==v$ has maximum zero. QED.

3.4 Full C++ solution

```
#include <bits/stdc++.h>
using namespace std;
int main(){ios::sync_with_stdio(false);cin.tie(nullptr);
    int n,q;if(!(cin>>n>>q)) return 0; int LOG=1; while((1<<LOG)<=n+1) LOG++;
    vector<vector<int>> up(LOG, vector<int>(n+1)), mx(LOG, vector<int>(n+1)); vector<int> depth
    for(int v=2;v<=n;v++){int p,w;cin>>p>>w; up[0][v]=p; mx[0][v]=w; depth[v]=depth[p]+1;}
    for(int k=1;k<LOG;k++) for(int v=1;v<=n;v++){int mid=up[k-1][v]; up[k][v]=up[k-1][mid]; mx
    auto lift=[&](int &v,int d){int ans=0; for(int k=0;k<LOG;k++) if(d&(1<<k)){ans=max(ans,mx[k
    while(q--){int a,b;cin>>a>>b; int ans=0; if(depth[a]<depth[b]) swap(a,b); ans=max(ans,lift
    }
```

3.5 Full Python solution

```

import sys
def main():
    data=list(map(int,sys.stdin.buffer.read().split()))
    if not data: return
    n,q=data[0],data[1]; idx=2; LOG=max(1,(n+1).bit_length())
    up=[[0]*(n+1) for _ in range(LOG)]; mx=[[0]*(n+1) for _ in range(LOG)]; depth=[0]*(n+1)
    for v in range(2,n+1):
        p,w=data[idx],data[idx+1]; idx+=2; up[0][v]=p; mx[0][v]=w; depth[v]=depth[p]+1
    for k in range(1,LOG):
        for v in range(1,n+1):
            mid=up[k-1][v]; up[k][v]=up[k-1][mid]; mx[k][v]=max(mx[k-1][v],mx[k-1][mid])
    def lift(v,d):
        ans=0; bit=0
        while d:
            if d&1: ans=max(ans,mx[bit][v]); v=up[bit][v]
            d//=2; bit+=1
        return v,ans
    out=[]
    for _ in range(q):
        a,b=data[idx],data[idx+1]; idx+=2; ans=0
        if depth[a]<depth[b]: a,b=b,a
        a,val=lift(a,depth[a]-depth[b]); ans=max(ans,val)
        if a!=b:
            for k in range(LOG-1,-1,-1):
                if up[k][a]!=up[k][b]:
                    ans=max(ans,mx[k][a],mx[k][b]); a=up[k][a]; b=up[k][b]
            ans=max(ans,mx[0][a],mx[0][b])
        out.append(str(ans))
    print("\n".join(out))
if __name__=="__main__": main()

```

3.6 Tests that matter

Test $u=v$, root and ancestor queries, a chain, maximum-edge boundaries, and both valid and invalid k values.

4 C. Kth Node On Path

4.1 Restatement

For query $u \ v \ k$, print the k -th node on the path from u to v , or -1 .

4.2 Observations, trace, and algorithm

Let $c=lca(u,v)$, $left=depth[u]-depth[c]+1$, and $total=depth[u]+depth[v]-2*depth[c]+1$. Reject invalid k . If $k \leq left$, return $lift(u,k-1)$; otherwise return $lift(v,total-k)$. On path 4,2,5 this returns 4,2,5 for $k=1,2,3$.

4.3 Correctness proof and complexity

The first left path nodes are exactly u -to- c ; the remaining nodes are v -to- c in reverse, so $total-k$ is the required upward distance from v . Time is $O(\log n)$ with the shared table. QED.

4.4 Full C++ solution

```
#include <bits/stdc++.h>
using namespace std;
int main(){ios::sync_with_stdio(false);cin.tie(nullptr);
    int n,q;if(!(cin>>n>>q)) return 0; int LOG=1; while((1<<LOG)<=n+1) LOG++;
    vector<vector<int>> up(LOG, vector<int>(n+1)); vector<int> depth(n+1);
    for(int v=2;v<=n;v++){cin>>up[0][v]; depth[v]=depth[up[0][v]]+1;}
    for(int k=1;k<LOG;k++) for(int v=1;v<=n;v++) up[k][v]=up[k-1][up[k-1][v]];
    auto lift=[&](int v,int d){for(int k=0;k<LOG;k++) if(d&(1<<k)) v=up[k][v]; return v;};
    auto lca=[&](int a,int b){if(depth[a]<depth[b]) swap(a,b); a=lift(a,depth[a]-depth[b]); if
    while(q--){int u,v,k;cin>>u>>v>>k; int c=lca(u,v); int left=depth[u]-depth[c]+1,total=depth
    }
```

4.5 Full Python solution

```
import sys
def main():
    data=list(map(int,sys.stdin.buffer.read().split()))
    if not data: return
```

```

n,q=data[0],data[1]; idx=2; LOG=max(1,(n+1).bit_length())
up=[[0]*(n+1) for _ in range(LOG)]; depth=[0]*(n+1)
for v in range(2,n+1):
    p=data[idx]; idx+=1; up[0][v]=p; depth[v]=depth[p]+1
for k in range(1,LOG):
    for v in range(1,n+1): up[k][v]=up[k-1][up[k-1][v]]
def lift(v,d):
    for k in range(LOG):
        if d>>k & 1: v=up[k][v]
    return v
def lca(a,b):
    if depth[a]<depth[b]: a,b=b,a
    a=lift(a,depth[a]-depth[b])
    if a==b: return a
    for k in range(LOG-1,-1,-1):
        if up[k][a]!=up[k][b]: a=up[k][a]; b=up[k][b]
    return up[0][a]
out=[]
for _ in range(q):
    u,v,k=data[idx],data[idx+1],data[idx+2]; idx+=3; c=lca(u,v); left=depth[u]-depth[c]+1
    if k<1 or k>total: out.append("-1")
    elif k<=left: out.append(str(lift(u,k-1)))
    else: out.append(str(lift(v,total-k)))
print("\n".join(out))
if __name__=="__main__": main()

```

4.6 Tests that matter

Test $u==v$, root and ancestor queries, a chain, maximum-edge boundaries, and both valid and invalid k values.

5 What to carry forward

6 Testing discussion

Run `CP_TARGET=solution python3 sections/41_euler_tour_lca_binary_lifting/check.py`. For random trees of at most 20 vertices, build a path by parent walks and use it as an oracle for all three exercises. This catches depth, aggregate-order, and one-based-rank mistakes more reliably than samples.

7 What to carry forward

Binary lifting turns an ancestor relation into a powers-of-two table. Once the exact jump invariant is established, LCA, path distance, path aggregates, and path indexing are small variations on the same controlled lifts.

Advanced algorithms become manageable when you state the invariant first, then code only the operations needed to preserve it.